

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 6

STRUCTURE OF ATMOSPHERE

- **STRATOSPHERE**

STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE



BENEFITS

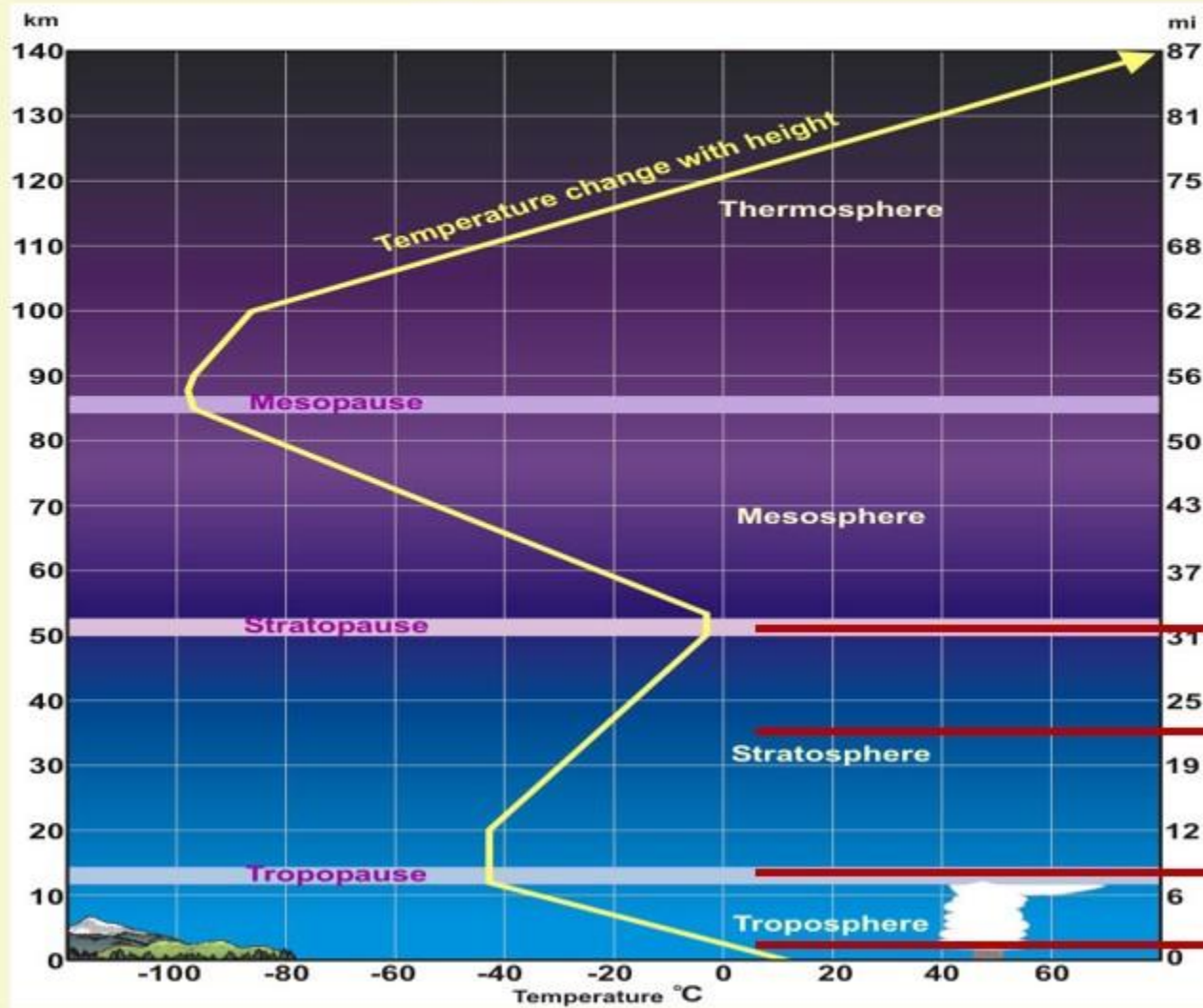
- Commercial jet planes fly here because of less atmospheric turbulence.
- Weather balloons are placed in this layer.

- 1 Stratosphere extends between 15 and 50 kilometre.
- 2 Temperature increases with increasing height.
- 3 Devoid of major weather phenomena but there is circulation of feeble winds and presence of cirrus cloud in the lower stratosphere.
- 4 Upper limit of the stratosphere is called as the stratopause.
- 5 Lower part consists of the most important layer- OZONE

STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE

TEMPERATURE PROFILE IN ATMOSPHERE LAYERS



→ Isothermal layer

→ Temp increases with height

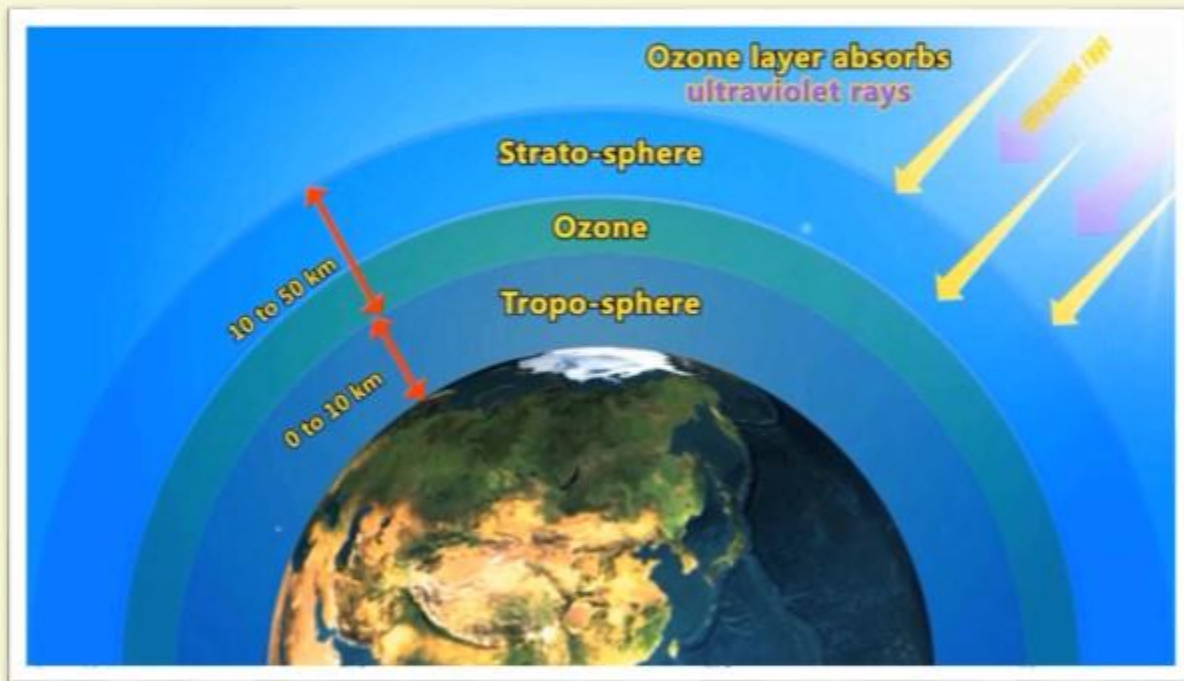
→ Isothermal layer

→ Temp decreases with height at NLR

STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE

OZONOSPHERE



1 Lower part of the Stratosphere is also known as ozonosphere.

2 Ozone is a faintly blue irritating gas with a characteristic pungent odour.

3 Protective cover for biological communities

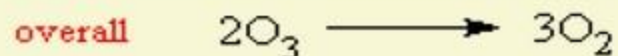
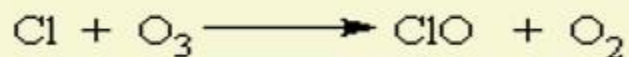
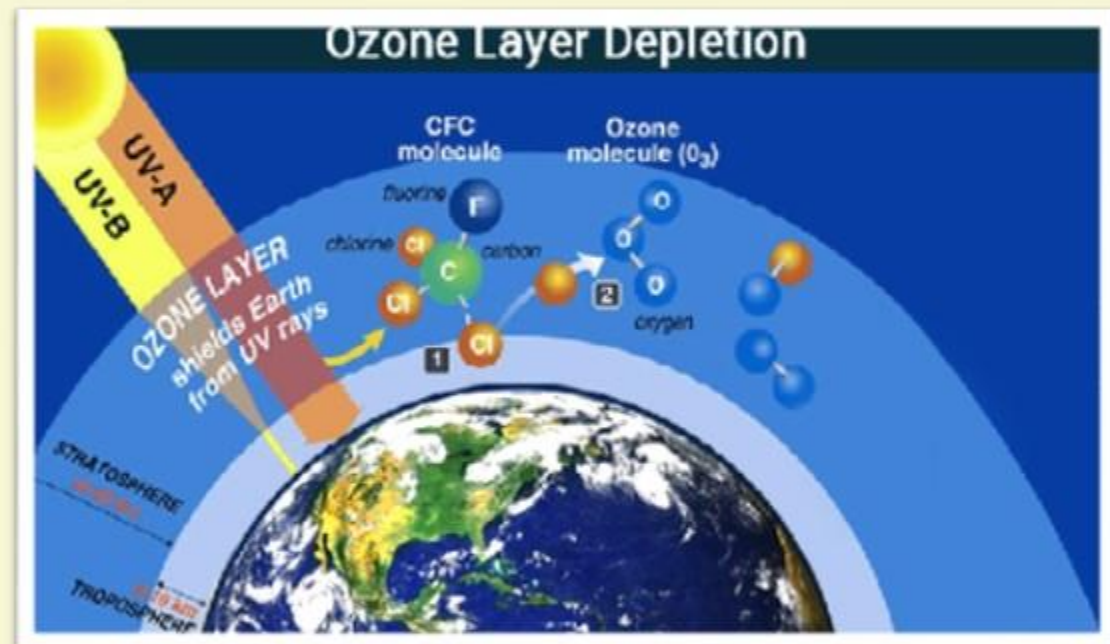
CHLOROFLUROCARBONS

They are the main culprits behind ozone depletion.

STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE

OZONOSPHERE



OZONE DEPLETION

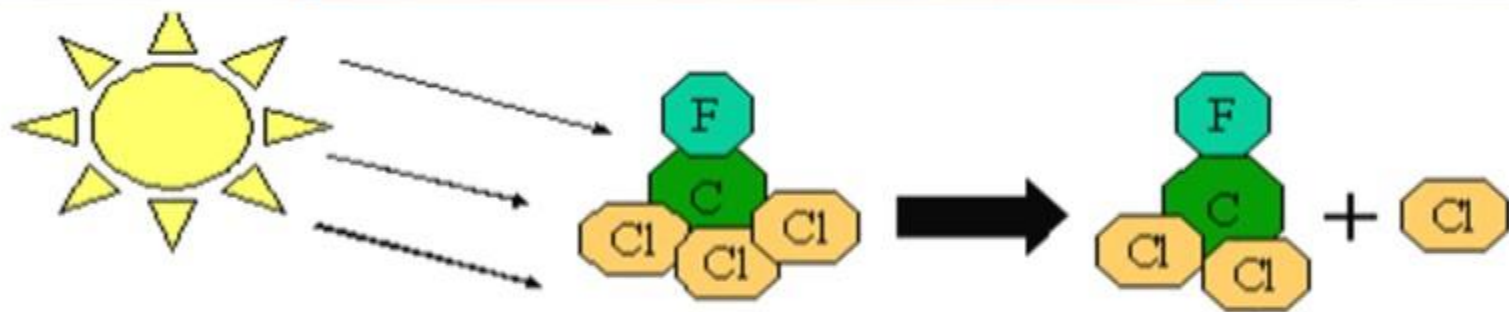
- 1 Chlorofluorocarbons, when used as propellants, are released into air and are transported into the stratosphere by vertical atmospheric circulation.
- 2 When the CFCs reach the upper atmosphere, high UV radiation reacts with these and releases chlorine atoms.
- 3 These free chlorine atoms react with ozone thereby leading to the depletion of the ozone layer
- 4 One chlorine atom can destroy millions of ozone atoms before it decomposes and this is why CFCs are extremely damaging to the ozone layer

STRUCTURE OF THE ATMOSPHERE

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OZONOSPHERE

PROCESS OF OZONE DEPLETION



UV rays strike CFC molecules, causing a Cl to break away



Lone Cl strikes ozone, leaving chlorine monoxide & oxygen molecule, which results in a loss of ozone

STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE

OZONOSPHERE

Consider the following statements:

Chlorofluorocarbons, known as ozone depleting substances, are used (UPSC Prelims 2012)

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statements given above is/are correct?

- [A] 1,2 and 3 only
- [B] 4 Only
- [C] 1,3 & 4 Only
- [D] 1,2,3 & 4

2 STRATOSPHERE OZONOSPHERE



OZONE-THE DOUBLE EDGED SWORD!



GOOD OZONE

Where is it found?

Present **naturally** in the
Stratosphere

Function:

It protects the Earth from harmful
UV radiation



BAD OZONE

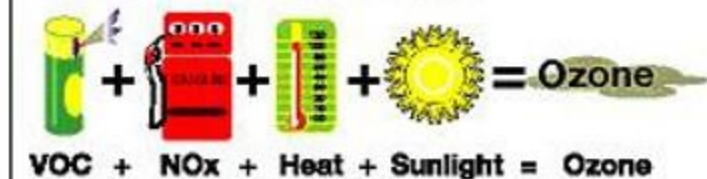
Where is it found?

Formed **anthropogenically** in the
Troposphere

Function:

- Does more harm than good
- Causes **lung problems**
- **Damages crops**
- Forms Photochemical Smog

The Ozone Cookbook:



STRUCTURE OF THE ATMOSPHERE

2 STRATOSPHERE

OZONOSPHERE

EXAMPLES OF VOCs



PHOTOCHEMICAL SMOG

1 Photochemical smog is the **chemical reaction** of **sunlight**, **nitrogen oxides** and **volatile organic compounds** in the atmosphere, which leaves airborne particles and ground-level ozone

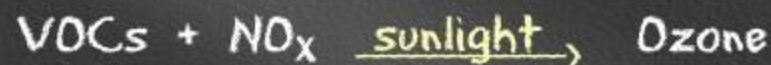
2 **Ground-level ozone** is a **secondary pollutant** that forms from the primary pollutants such as Volatile Organic Compounds (Hydrocarbons) and nitrogen oxides (NO_x) in the presence of sunlight

PRIMARY POLLUTANT

A primary pollutant is an air pollutant emitted directly from a source

SECONDARY POLLUTANT

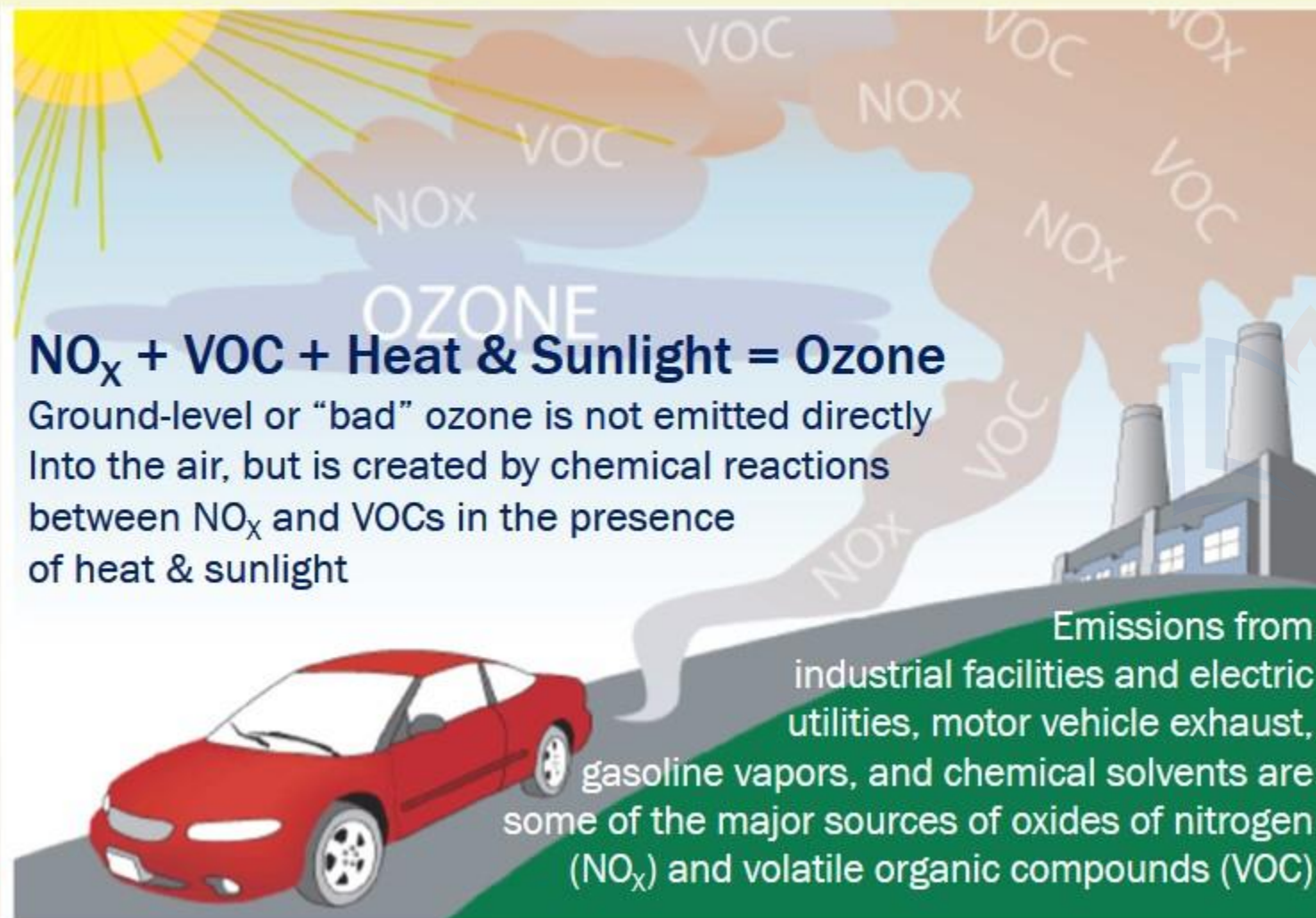
A secondary pollutant is not directly emitted as such, but forms when other primary pollutants react in the atmosphere.



STRUCTURE OF THE ATMOSPHERE

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OZONOSPHERE



PHOTOCHEMICAL SMOG

- 1 Photochemical smog is therefore considered to be **a problem of modern industrialization.**
- 2 It is present in all modern cities, but it is **more common in cities with sunny, warm, dry climates** and a large number of motor vehicles. (Warm upper air will inhibit vertical circulation of winds because of a phenomena known as temperature inversion)

POSSIBLE MAINS QUESTION

TOPIC:

GS Paper I : Urbanization- their problems and remedies

GS Paper III : Environmental pollution and degradation

SOURCE:

THE HINDU- The much neglected pollutant

What is ground-level ozone? Why is it regarded as the “much neglected pollutant”?

HINTS

1 Definition of ground level ozone/ bad ozone/ tropospheric ozone

2 Secondary pollutant

3 VOC + NO_x + sunlight

4 Main ingredient of photochemical smog

5 Absence of government policy effort

6 Not much attention given when compared to other pollutants

7 International Conventions like Montreal concentrate only on ODS, stratospheric ozone and not on ground-level ozone.

8 Lack of awareness on its harmfulness.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

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